

TTDS Coursework 2 Report

IR Evaluation, Text Analysis, and Classification

Student ID: s2414220

December 2, 2025

1 Implementation Overview

This coursework implements IR evaluation metrics, text corpus analysis (MI, χ^2 , LDA), and sentiment classification in Python. Code is organized into three classes: `IREvaluator`, `TextAnalyzer`, and `SentimentClassifier`. Key challenges included: (1) correctly implementing nDCG with Lecture 9's formula ($\text{rel}_i / \log_2(i)$ for $i \geq 2$), (2) addressing Chi-square's negative correlation issue through positive association filtering, and (3) balancing model complexity in classification to avoid overfitting. The work provided experience with metric implementation, statistical testing, feature selection, topic modeling, and error-driven improvement.

2 IR Evaluation Results

2.1 System Performance and Statistical Significance

Table 1 presents mean performance and significance testing. System 3 achieves highest scores in 5/6 metrics (all $p < 0.05$).

Table 1: IR System Performance and Statistical Significance

Sys	P@10	R@50	R-P	AP	nDCG@10	nDCG@20	Best?	p-val
1	0.390	0.834	0.401	0.400	0.363	0.485	-	-
2	0.220	0.867	0.252	0.300	0.200	0.246	R@50	0.703
3	0.410	0.767	0.449	0.451	0.420	0.511	5/6	< 0.86
4	0.080	0.189	0.049	0.075	0.069	0.076	-	-
5	0.410	0.767	0.358	0.364	0.332	0.424	tie P@10	1.0
6	0.410	0.767	0.449	0.445	0.400	0.490	tie R-P	1.0

2.2 Analysis and Discussion

System 3 dominates 5/6 metrics but lacks statistical significance (all $p > 0.05$) due to small sample size ($n=10$ queries). Limited queries yield high variance, obscuring differences. Additionally, tied scores (Systems 3, 5, 6 all scored 0.410 on P@10) produce $p=1.000$. Despite null results, System 3's consistent superiority suggests practical advantage. System 2's R@50 superiority reflects a recall-focused strategy trading precision for retrieval breadth. The lack of significance reflects test collection limitations rather than absence of real performance differences.

3 Text Analysis

3.1 Corpus Overview

The analysis focused on three religious text corpora: the Quran, Old Testament (OT), and New Testament (NT). Each verse was treated as a separate document. Preprocessing included tokenization, lowercasing, and stopword removal.

Table 2: Top 5 Features by MI and χ^2 for Each Corpus

Method	Quran	Old Testament	New Testament
MI	bargain (2.576), trunks, mim, needlessly, vicious	overflows (0.690), circumference, ishpan, embalm, dismayed	eunice (2.236), infallible, bethphage, rigid, apelles
χ^2	muhammad (1852), god (1793), certainly (1683), believers (1588), torment (1382)	shall (1505), lord (1114), israel (1096), king (862), land (472)	jesus (3027), christ (1764), disciples (741), things (674), paul (529)

3.2 Mutual Information and Chi-Square Analysis

3.2.1 Top Features by MI and χ^2

3.2.2 Comparison of MI and χ^2 Rankings

MI identifies corpus-exclusive terms (all top words score identically: $\log_2(N/N_c)$) but favors rare vocabulary with limited interpretability. χ^2 highlights high-frequency discriminators that capture thematic identity: “muhammad/believers” (Quran theology), “israel/king/david” (Hebrew monarchy), “jesus/christ/disciples” (Christian Gospel). MI measures lexical uniqueness (perfect discrimination even for singletons), while χ^2 tests statistically significant deviation, favoring frequent terms. For corpus characterization, χ^2 proves more semantically meaningful.

3.3 LDA Topic Modeling

An LDA model with 20 topics was trained on all verses from the three corpora. For each corpus, the most prominent topic (highest average document-topic probability) was identified.

Table 3: Most Prominent Topics and Top Tokens for Each Corpus

Corpus	Topic ID (Avg Score)	Top 10 Tokens
Quran	Topic 19 (0.358)	god, people, would, say, one, muhammad, certainly, lord, torment, know
Old Testament	Topic 16 (0.080)	king, came, david, saying, said, sent, lord, people, jerusalem, house
New Testament	Topic 10 (0.143)	things, jesus, said, life, christ, answered, god, world, one, come

3.3.1 Topic Labels

Based on the top tokens, I assign the following interpretive labels to these topics:

- **Quran Topic 19:** “Divine Guidance and Prophethood” — dominated by theological terminology (god, lord, muhammad, torment) reflecting Islamic monotheism and prophetic teachings
- **Old Testament Topic 16:** “Davidic Monarchy and Temple” — centered on Hebrew kingship (king, david, jerusalem, house) representing the historical-political narrative of ancient Israel
- **New Testament Topic 10:** “Christ’s Life and Teachings” — focused on Jesus’s ministry (jesus, christ, life, things, world) capturing the Gospel narrative and theological discourse

3.3.2 LDA Insights and Comparison with MI/ χ^2

LDA reveals thematic coherence differences: Quran (0.358) shows unified theology, NT (0.143) balances Gospel with epistles, OT (0.080) exhibits diverse genres. Topic 19 (Quran-specific, 0.358 vs 0.018 in OT) and Topic 10 (NT-dominant, 0.143 vs 0.024 in OT) demonstrate distinct theological identities. Topics 5-7 show OT/NT continuity absent in Quran.

MI/ χ^2 identify discriminative words (“muhammad” vs “jesus”), while LDA reveals co-occurring patterns (“king+david+jerusalem”). LDA’s key advantage: capturing polysemy (“lord” has different contexts across corpora) through topic distributions, whereas MI/ χ^2 treat words atomically.

4 Text Classification

4.1 Dataset and Experimental Setup

The sentiment classification task involved three-way classification (positive, negative, neutral) of tweets. The dataset comprised 18,646 training samples with the following distribution:

- Neutral: 8,789 (47.1%)
- Positive: 5,979 (32.1%)
- Negative: 3,878 (20.8%)

The data was shuffled and split 90/10 into training (16,781) and development (1,865) sets using stratified sampling to maintain class proportions. All experiments used random seed 42 for reproducibility. A separate test set of 4,662 samples was provided for final evaluation.

4.2 Baseline System

The baseline system followed the standard approach:

- **Preprocessing:** Lowercasing, basic tokenization
- **Features:** Bag-of-words (BOW) with `CountVectorizer` (`max_features=10,000`, `min_df=2`)
- **Classifier:** Linear SVM with $C = 1000$, trained using `LinearSVC`

Table 4: Baseline Performance (F1 Scores Only)

Split	F-Pos	F-Neg	F-Neu	F-Macro	Acc
Train	0.996	0.999	0.997	0.997	0.997
Dev	0.545	0.446	0.561	0.518	0.532
Test	0.553	0.449	0.560	0.520	0.535

Baseline shows severe overfitting (Train 0.997 vs Dev 0.518) but consistent dev/test performance (0.518 vs 0.520), validating split quality. Negative class worst (0.446) due to fewest samples.

4.3 Error Analysis

Three misclassifications reveal common challenges:

1. **Political news:** “Wikileaks: Clinton Foundation...” (Pred: neutral, Act: negative). Factual style masks negative implications; BOW cannot capture “Raises Questions” as implicit criticism.
2. **Positive keywords:** “Bob Dylan... all star lineup...” (Pred: positive, Act: neutral). “All star” triggers false positive despite event announcement being informational.
3. **Context-dependent:** “Iran islamist revolution... blinded by fact’s clarity” (Pred: positive, Act: negative). “Clarity” interpreted positively; model misses “blinded by” criticism.

4.4 Improved System

Four targeted improvements address error analysis findings:

1. **TF-IDF:** Down-weights common words, up-weights discriminative terms (`sublinear_tf=True`, `max_df=0.95`). Improved precision distinguishing neutral from positive/negative.
2. **Bigrams:** Captures negation (“not good”) and multi-word expressions (`ngram_range=(1,2)`). Negative class F1: 0.446→0.500 (+12%).
3. **C=500:** Reduced from 1000 to mitigate overfitting. Better generalization while maintaining high training performance.
4. **20K features:** Expanded from 10K to accommodate bigrams, enabling richer context representation.

Table 5: Improved Performance (F1) and Gains vs Baseline

Split	F-Pos	F-Neg	F-Neu	F-Macro	Acc	Gain
Train	0.999	0.999	0.999	0.999	0.999	+0.002
Dev	0.581	0.500	0.589	0.557	0.568	+0.039
Test	0.594	0.503	0.596	0.564	0.576	+0.044

4.5 Performance Analysis

Improvements achieved +7.5% (dev) and +8.5% (test) gains. Per-class: positive (+6.6%/+7.4%), negative (+12.1%/+12.0%), neutral (+5.0%/+6.4%). Negative class’s substantial gain validates error analysis hypothesis: bigrams effectively capture negation patterns.

Test performance (0.564) slightly exceeds dev (0.557, +0.007), indicating: (1) no overfitting to dev set—hyperparameters generalize well, (2) representative 90/10 stratified split maintained distribution, (3) robust TF-IDF+bigram features transfer across partitions. Consistent dev/test performance (1% deviation) suggests model maintains 0.56 F1 on similar data.

5 Conclusion

This coursework demonstrated: (1) IR evaluation metrics implementation with significance testing revealing small sample limitations ($n=10$), (2) complementary feature selection methods (MI: exclusive vocabulary, χ^2 : high-frequency discriminators, LDA: thematic structure) capturing religious texts’ nature quantitatively (Quran coherence: 0.358, NT: 0.143, OT: 0.080), and (3) error-driven classification improvement (+8.5% gain) validating targeted feature engineering (TF-IDF+bigrams).

Key lesson: computational text analysis requires understanding algorithms’ assumptions and limitations, not just implementation. Main challenges: nDCG formula correctness, Chi-square correlation handling, overfitting mitigation. Future directions: larger query sets for statistical power, hierarchical LDA for sub-themes, transformer-based sentiment classifiers.